

3.1 Defining Features as Representational Primitives

Features as fundamental units of representation. The notion of a *feature* in neural networks is central yet elusive, reflecting the pre-paradigmatic state of mechanistic interpretability. We adopt the notion of *features* as the *fundamental units of neural network representations*, such that features cannot be further *disentangled* into simpler, distinct factors. These features are core components of a neural network’s representation, analogous to how cells form the fundamental unit of biological organisms (Olah et al., 2020).

Definition 1: Feature

Features are the fundamental units of neural network representations that cannot be further decomposed into simpler independent factors.

Concepts as natural abstractions. The world consists of various entities that can be grouped into categories or *concepts* based on shared properties. These concepts form high-level summaries like "tree" or "velocity," allowing compact world representations by discarding many irrelevant low-level details. Neural networks can capture and represent such *natural abstractions* (Chan et al., 2023) through their learned *features*, which serve as building blocks of their internal representations, aiming to capture the *concepts* underlying the data.

Features encoding input patterns. In traditional machine learning, *features* are understood as characteristics or attributes derived directly from the input data stream (Bishop, 2006). This view is particularly relevant for systems focused on *perception*, where features map closely to the input data. However, in more advanced systems capable of *reasoning* with abstractions, features may emerge internally within the model as representational patterns, even when processing information unrelated to the input. In this context, features are better conceptualized as *any measurable property or characteristic of a phenomenon* (Olah, 2022), encoding abstract concepts rather than strictly reflecting input attributes.

Features as representational atoms. A key property of features is their irreducibility, meaning they cannot be decomposed into or expressed as a combination of simpler, independent factors. In the context of input-related features, Engels et al. (2024) define a feature as *irreducible* if it cannot be decomposed into or expressed as a combination of statistically independent patterns or factors in the original input data. Specifically, a feature is reducible if transformations reveal its underlying pattern, which can be separated into independent co-occurring patterns or is a mixture of patterns that never co-occur. We propose generalizing this notion of irreducibility to features encoding abstract concepts not directly tied to input patterns, such that features cannot be reduced to combinations or mixtures of other independent components within the model’s representations.

Features beyond human interpretability. Features could be defined from a *human-centric perspective* as *semantically meaningful, articulable input patterns encoded in the network’s activation space* (Olah, 2022). However, while cognitive systems may converge on similar *natural abstractions* (Chan et al., 2023), these need not necessarily align with human-interpretable *concepts*. Adversarial examples have been interpreted as non-interpretable features meaningful to models but not humans. Imperceptible perturbations fool networks, suggesting reliance on alien representational patterns (Ilyas et al., 2019). As models surpass human capabilities, their learned features may become increasingly abstract, encoding information in ways incongruent with human intuition (Hubinger, 2019a). Mechanistic interpretability aims to uncover the *actual* representations learned, even if diverging from human concepts. While human-interpretable concepts provide guidance, a non-human-centric perspective that defines features as independent model components, whether aligned with human concepts or not, is a more comprehensive and future-proof approach.

3.2 Nature of Features: From Monosemantic Neurons to Non-Linear Representations

Neurons as Computational Units? In the architecture of neural networks, neurons are the natural computational units, potentially representing individual features. Within a neural network representation

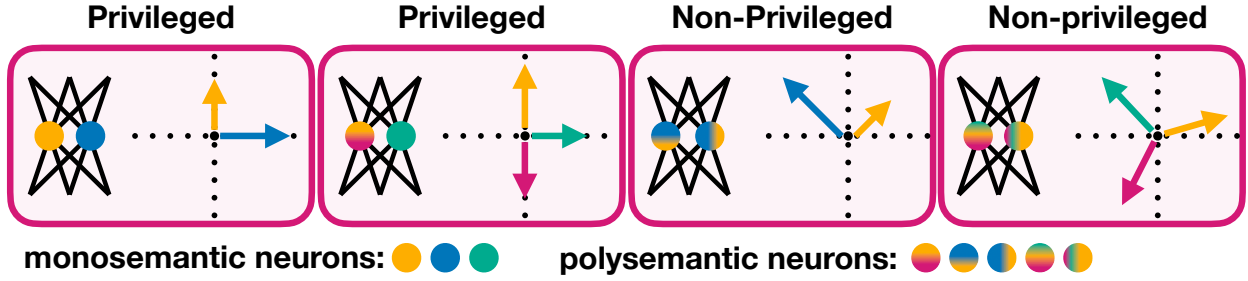


Figure 3: Contrasting privileged and non-privileged bases. In a non-privileged basis, there is no reason to expect features to be basis-aligned – calling basis dimensions neurons has no meaning. In a privileged basis, the architecture treats basis directions differently – features can but need not align with neurons (Bricken et al., 2023). **Leftmost:** Privileged basis; individual features (arrows) align with basis directions, resulting in *monosemantic* neurons (colored circles). **Middle left:** Privileged basis, where despite having more features than neurons, some neurons are monosemantic, representing individual features, while others are *polysemantic* (overlapping gradients), encoding *superposition* of multiple features. **Middle right:** Non-privileged basis where, even when the number of features equals the number of neurons, the lack of alignment between the feature directions and basis directions results in polysemantic neurons encoding combinations of features. **Rightmost:** Non-privileged, polysemantic neurons as feature directions do not align with neuron basis.

$h \in \mathbb{R}^n$, the n basis directions are called neurons. For a neuron to be meaningful, the basis directions must functionally differ from other directions in the representation, forming a *privileged basis* – where the basis vectors are architecturally distinguished within the neural network layer from arbitrary directions in activation space, as shown in Figure 3. Typical non-linear activation functions privilege the basis directions formed by the neurons, making it meaningful to analyze individual neurons (Elhage et al., 2022b). Analyzing neurons can give insights into a network’s functionality (Sajjad et al., 2022; Mu & Andreas, 2020; Dai et al., 2022; Ghorbani & Zou, 2020; Voita et al., 2023; Durrani et al., 2020; Goh et al., 2021; Bills et al., 2023; Huang et al., 2023).

Monosemantic and Polysemantic Neurons. A neuron corresponding to a single semantic concept is called *monosemantic*. The intuition behind this term comes from analyzing what inputs activate a given neuron, revealing its associated semantic meaning or concept. If neurons were the representational primitives of neural networks, all neurons would be monosemantic, implying a one-to-one relationship between neurons and features. Comprehensive interpretability would be as tractable as characterizing all neurons and their connections. However, empirically, especially for transformer models (Elhage et al., 2022b), neurons are often observed to be *polysemantic*, *i.e.*, associated with multiple, unrelated concepts (Arora et al., 2018; Mu & Andreas, 2020; Elhage et al., 2022a; Olah et al., 2020). For example, a single neuron may be activated by both images of cats and images of cars, suggesting it encodes multiple unrelated concepts. Polysemanticity contradicts the interpretation of neurons as representational primitives and, in practice, makes it challenging to understand the information processing of neural networks.

Exploring Polysemanticity: Hypotheses and Implications. To understand the widespread occurrence of polysemanticity in neural networks, several hypotheses have been proposed:

- i.) One trivial scenario would be that feature directions are orthogonal but not aligned with the basis directions (neurons). There is no inherent reason to assume that features would align with neurons in a non-privileged basis, where the basis vectors are not architecturally distinguished. However, even in a privileged basis formed by the neurons, the network could represent features not in the standard basis but as linear combinations of neurons (see Figure 3, middle right).
- ii.) An alternative hypothesis posits that *redundancy due to noise* introduced during training, such as random dropout (Srivastava et al., 2014), can lead to redundant representations and, consequently,

to polysemantic neurons (Marshall & Kirchner, 2024). This process involves distributing a single feature across several neurons rather than isolating it into individual ones, thereby encouraging polysemanticity.

- iii.) Finally, the *superposition* hypothesis addresses the limitations in the network’s representative capacity – the number of neurons versus the number of crucial concepts. This hypothesis argues that the limited number of neurons compared to the vast array of important concepts necessitates a form of compression. As a result, an n -dimensional representation may encode features not with the n basis directions (neurons) but with the $\propto \exp(n)$ possible almost orthogonal directions (Elhage et al., 2022b), leading to polysemanticity.

Hypothesis 1: Superposition

Neural networks represent more features than they have neurons by encoding features in overlapping combinations of neurons.

Superposition Hypothesis. The *superposition* hypothesis suggests that neural networks can leverage high-dimensional spaces to represent more features than the actual count of neurons by encoding features in almost orthogonal directions. Non-orthogonality means that features interfere with one another. However, the benefit of representing many more features than neurons may outweigh the interference cost, mainly when concepts are sparse and non-linear activation functions can error-correct noise (Elhage et al., 2022b).

Toy models can demonstrate under which conditions superposition occurs (Elhage et al., 2022b; Scherlis et al., 2023). Neural networks, via superposition, may effectively simulate computation with more neurons than they possess by allocating each feature to a linear combination of neurons, creating what is known as an overcomplete linear basis in the representation space. This perspective on superposition suggests that polysemantic models could be seen as compressed versions of hypothetically larger neural networks where each neuron represents a single concept (see Figure 5). Consequently, an alternative definition of features could be:

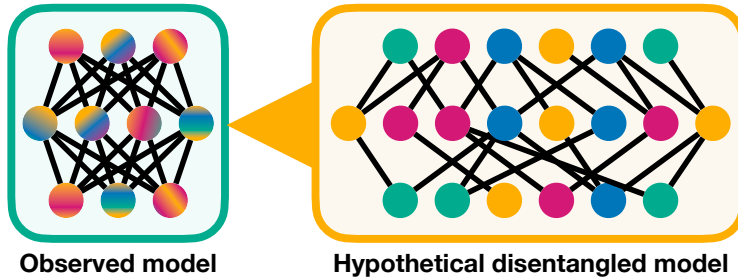


Figure 5: Observed neural networks (left) can be viewed as compressed simulations of larger, sparser networks (right) where neurons represent distinct features. An "almost orthogonal" projection compresses the high-dimensional sparse representation, manifesting as polysemantic neurons involved with multiple features in the lower-dimensional observed model, reflecting the compressed encoding. Figure adapted from (Bricken et al., 2023).

Research on superposition, including works by (Elhage et al., 2022b; Scherlis et al., 2023; Henighan et al., 2023), often investigates simplified models. However, understanding superposition in practical, transformer-based scenarios is crucial for real-world applications, as pioneered by Gurnee et al. (2023).

The need for understanding networks despite polysemanticity has led to various approaches: One involves training models without superposition (Jermyn et al., 2022), for example, using a softmax linear unit (Elhage et al., 2022a) as an activation function to empirically increase the number of *monosemantic* neurons, but at the cost of making other neurons less interpretable. From a capabilities standpoint, polysemanticity may